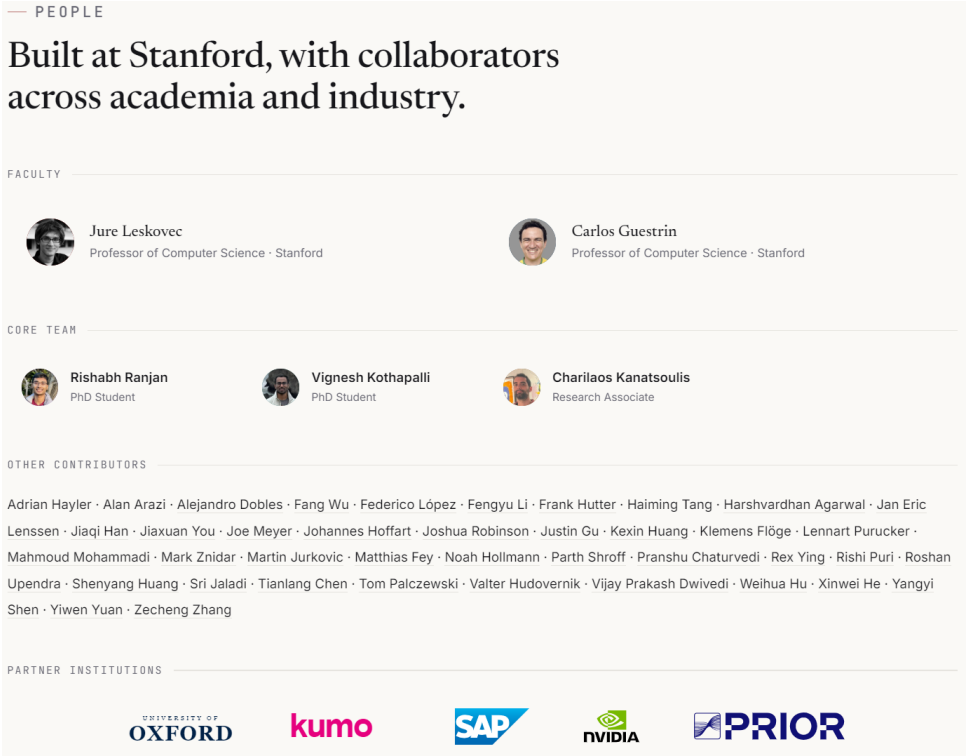
Core Content

- **RelBench: A Benchmark for Deep Learning on Relational Databases**
  - Code https://github.com/stanford-star/relbench
  - Huggingface https://huggingface.co/stanford-star
  - Paper https://arxiv.org/pdf/2407.20060?



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DDL: Rel-event Relational Database Schema

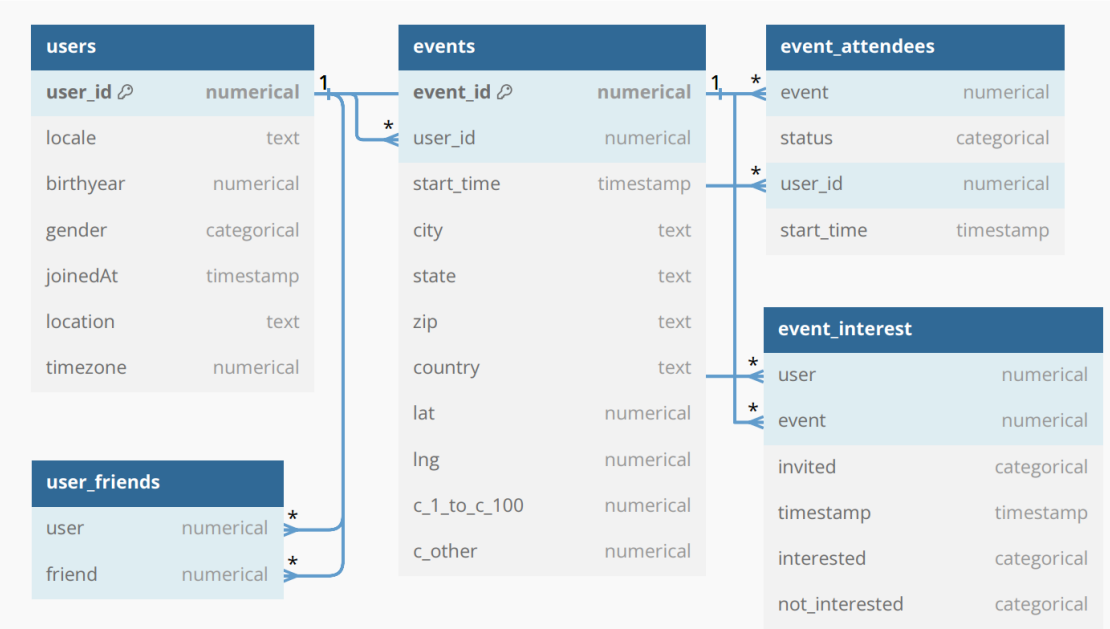
☐ Details

Blue Background : Primary Key
- *Except events.user_id*
- *Null not allowed*

Categorical : numerical (0 or 1)

1 : One-side, anchored to the Primary Key

***** : Many-side, marks the Foreign Key



DDL: Open ddl_practice.py

- ☐ **Write SQL between the quotes after the PRAGMA statement.**

```python
import sqlite3


# Write your CREATE TABLE statements between the quotes.
# Do not delete the PRAGMA statement. Write your CREATE TABLE statements after the PRAGMA statement.
DDL = r"""
PRAGMA foreign_keys = ON;

"""

# Do not modify the test data below. It will be used to test your DDL statements.

TEST_DATA = r"""
PRAGMA foreign_keys = ON;
```

Below is the test code. Run it to check your implementation.

Take a screenshot showing your source code and result, then add them to your submission folder.

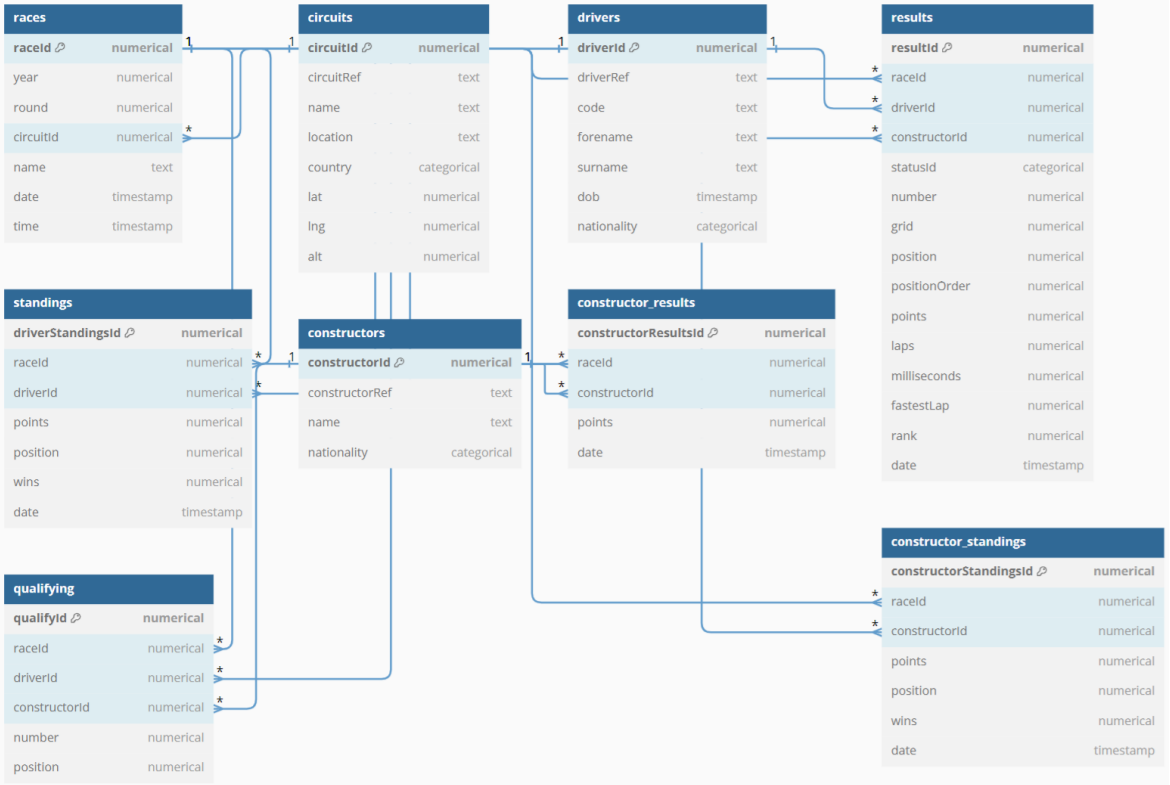
Rel-f1 Relational Database Schema

☐ Details

🔑 : Primary Key

1 : One-side, anchored to the Primary Key

***** : Many-side, points to a Foreign Key



Step 1: Open NL_queries.txt

☐ Seven Natural Language Queries

- (Q1) **Easy** : Selection and Ordering
- (Q2) **Easy** : JOIN
- (Q3) **Easy** : GROUP BY + COUNT + HAVING
- (Q4) **Medium** : JOIN + multiple aggregates
- (Q5) **Medium** : JOIN + multiple aggregates
- (Q6) **Hard** : Nested query + Multi-condition Filtering
- (Q7) **Hard** : JOIN + GROUP BY + COUNT + HAVING

Step 1: Open NL_queries.txt

- **Seven Natural Language Queries**
  - (Q1) **Easy** : Selection and Ordering

> Retrieve the forename, surname, and nationality of all British drivers.
> Sort the result by surname and then forename in ascending order.

Step 1: Open NL_queries.txt

- **Seven Natural Language Queries**
  - (Q2) **Easy** : JOIN

> Retrieve the race name, year, and circuit name for all races held at circuits in Italy. Sort the result by year in ascending order.

Step 1: Open NL_queries.txt

- **Seven Natural Language Queries**
  - (Q3) **Easy** : GROUP BY + COUNT + HAVING

> For each constructor, retrieve the constructor name and the number of race results associated with that constructor. Show only constructors having at least 20 race results.

Step 1: Open NL_queries.txt

- **Seven Natural Language Queries**
  - (Q4) **Medium** : JOIN + multiple aggregates

> For each driver who has at least 20 race results, retrieve the driver's first name, last name, number of race results, and average points per race result. Show only drivers whose average points are greater than 5.

Step 1: Open NL_queries.txt

- **Seven Natural Language Queries**
  - (Q5) **Medium** : JOIN + multiple aggregates

> A sports journalist claims: “Ferrari has never achieved 15 or more podium finishes in a single season.” Write an SQL query that can be used to check whether this claim is false. For each piece of evidence, report the season (year) and the relevant statistic.

Step 1: Open NL_queries.txt

- **Seven Natural Language Queries**
  - (Q6) **Hard** : Nested query + Multi-condition Filtering

> A Formula 1 fan argues: “Winning in Italy and winning in the United Kingdom require very different strengths. I doubt many drivers have managed to win races in both countries.” Write an SQL query to find drivers who have won at least one race in Italy and at least one race in the United Kingdom. Report the driver's first and last name.

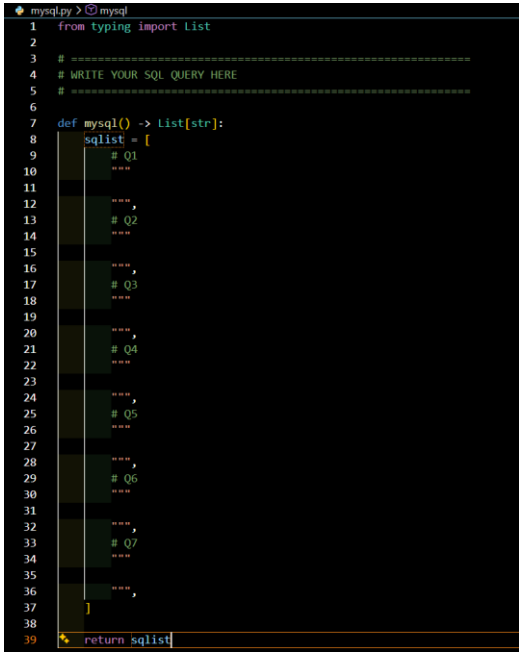
Step 1: Open NL_queries.txt

- **Seven Natural Language Queries**
  - (Q7) **Hard** : JOIN + GROUP BY + COUNT + HAVING

> Mike is a big fan of Max Verstappen. Emily teases him by saying, "Max Verstappen has never won more than 10 races in a single season." Mike believes that Emily is wrong. Write an SQL query that Mike can use to find evidence in the database that disproves Emily's claim. For each piece of evidence, report the season (year) and the relevant statistic.

Step 2: Open mysql.py

- ☐ **Write SQL to Previous Seven Questions**



Important: SQLite doesn't support ALL or ANY

Step 3: Run test.py

Procedures to Check Results

- First change *“my_student_id”* in test.py to your id
- Type *“python test.py”* in terminal
- A folder containing retrieved results is produced

Step 4: Submit

Compress the Folder as Zip

- Submit to Eclass
- Be sure to check that all the CSV files and the screenshot of the DDL are included

